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# A Nano-Micro Hybrid Structure Composed of $Fe_7S_8$ Nanoparticles Embedded in Nitrogen-Doped Porous Carbon Framework for High-Performance Lithium/Sodium-Ion Batteries

*Zhiming Liu, Fang Hu, Juan Xiang, Chuang Yue, Dongsoo Lee, and Taeseup Song**

**Iron sulfides are attractive anode materials for lithium-ion batteries (LIBs) and sodium-ion batteries (SIBs) due to their high theoretical capacities, low cost, and eco-friendliness. However, their real application is greatly hindered by the rapid capacity fading caused by the large volume changes and sluggish kinetics of iron sulfides during the charge and discharge processes. Combining with carbonaceous materials and tuning the structure at nanoscale are essential to address this issue. Here, a facile hydrothermal method coupled with a carbonization process is developed to synthesize a nano-micro hybrid porous structure, which is composed of $Fe_7S_8$ nanoparticles embedded in nitrogen-doped carbon framework ($Fe_7S_8$@NC-PS). This hierarchical sphere is constructed by interconnected 2D nanowalls. The as-prepared $Fe_7S_8$@NC-PS electrodes reveal excellent rate capability and cycling stability in LIBs and SIBs. The remarkable electrochemical properties are attributed to the porous nano-micro hybrid architecture and the high conductivity and structural stability of the nitrogen-doped carbon framework.**

High-performance rechargeable batteries are increasingly demanded for applications in portable electronic devices, electric vehicles (EV), and stationary storage systems. Lithium-ion batteries (LIBs) have been successfully commercialized in portable electronics and EV owing to the high energy density.[1–3] As a promising alternative to LIBs, sodium-ion batteries (SIBs) have attracted increasing attention because of the natural abundance and environmental benignity of sodium resources.[4–6] Finding appropriate electrode materials with high electrochemical performance and low cost is still a main challenge currently. Among all the electrode materials, iron sulfides have been considered as promising anode materials for LIBs and SIBs due to their high theoretical capacities, abundant resources, and eco-friendly.[7–14] However, iron sulfides show poor electrochemical performance owing to their intrinsic low conductivity, sluggish kinetics, and large volume changes during the charging and discharging processes, which hinders their practical application.[15–17]

Combining with carbonaceous materials is a feasible approach to alleviate these issues. Carbon not only acts as a buffer to mitigate pulverization, but also works as a conductive matrix during cycling.[16,18] Constructing active materials into nanostructures has been considered another effective strategy.[19,20] Generally, materials with nanostructures can alleviate mechanical stress without the initiation of fractures and enhance the reaction kinetics by reducing the transport length for electrons and ions during cycling, thereby improving the electrochemical performance. However, nanostructured materials usually suffer from a low-tap density because of the large interparticle space and high surface area.[21,22] Besides, the large contact area with electrolyte leads to excess formation of solid electrolyte interface (SEI) layer.[21,23] These problems greatly compromise the merits of nanostructured electrode materials over bulk materials. Engineering a nano-micro hierarchical structure has been considered a promising strategy.[24–27] These unique architectures show high cycling stability and superior rate capability benefiting from the nanoscale structure, while remaining the advantages of packing density and SEI formation provided by the microscale dimension.

Herein, we have successfully fabricated a unique nano-micro hierarchical structure organized by $Fe_7S_8$ nanoparticles embedded in nitrogen-doped carbon porous spheres through a facile hydrothermal method coupled with a carbonization process ($Fe_7S_8$@NC-PS). The porous carbon framework was constructed by interconnected 2D nanowalls. This porous structure offered sufficient void space for volume changes during cycling without causing serious pulverization. Moreover, 2D nanowalls provided a large contact area with electrolyte and short diffusion path for electrons and ions, which significantly facilitated the reaction kinetics.[28–30] In addition, the nitrogen-doped carbon framework contributed a high conductivity and structural stability upon charge and discharge.[31–33] Generally, the well-bonded nitrogen atom could drastically change the

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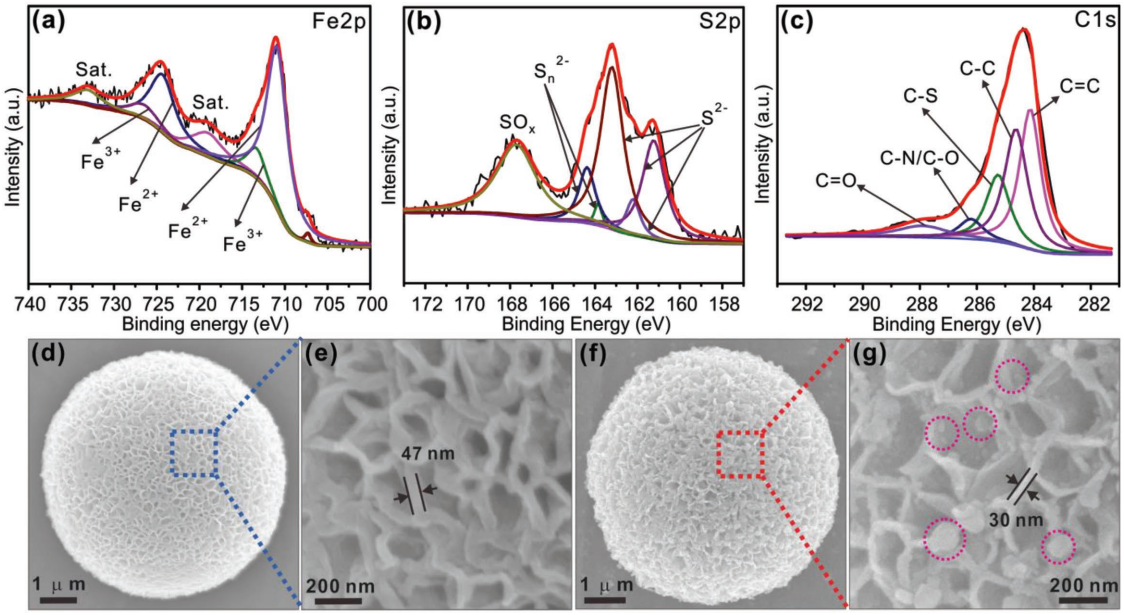


**Figure 1.** XPS spectra of $Fe_7S_8$@NC-PS: a) high-resolution Fe 2p XPS spectrum, b) S 2p XPS spectrum, and c) C 1s spectrum. FESEM images of d,e) $Fe_7S_8$@NC-PS precursor without carbonization and f,g) $Fe_7S_8$@NC-PS.

electronic performance, provide more active sites, and enhance the interaction between the carbon structure and Li/Na ions, thus it is expected to improve the kinetics of Li/Na diffusion and transfer.[34] Besides, N-doping is capable of increasing the specific capacity as a result of more favorable binding of N-doped sites with Li/Na ions.[34,35] Benefiting from these structural and composition merits, $Fe_7S_8$@NC-PS revealed a high specific capacity, excellent rate capability, and long cycling stability in LIBs and SIBs.

Figure S1a (Supporting Information) shows the X-ray diffraction (XRD) patterns of $Fe_7S_8$@NC-PS precursor without carbonization. All peaks were mainly assigned to the $FeS_2$ phase (JCPDS No. 42-1340). After annealing at 500 °C in Ar, $FeS_2$ totally transformed into $Fe_7S_8$ phase without impurities, which was confirmed by XRD measurement (Figure S1b, Supporting Information, JCPDS No. 24-0220). An X-ray photoelectron spectra (XPS) technique was applied to investigate the composition and valence states of $Fe_7S_8$@NC-PS. The survey spectrum indicated the presence of Fe, S, C, O, and N elements (Figure S2, Supporting Information). The deconvolution analysis then focused on the Fe 2p, S 2p, and C 1s regions. **Figure 1**a shows the high-resolution Fe 2p XPS spectrum. The peak located at 707.3 eV was attributed to $Fe^0$; the peaks centered at 710.8 and 724.3 eV indicated to $Fe^{2+}$ valence state; whereas the peaks at 713.6 and 726.6 eV were assigned to $Fe^{3+}$.[16,36,37] In the high-resolution S 2p XPS spectrum (Figure 1b), peaks at 161.5, 162.8, 163.5, 164.1, and 165.3 eV corresponded to $S^{2-}$ and $S_n^{2-}$. The other peak at 168.6 eV was assigned to the oxidized sulfur species (e.g., $SO_x$).[16,36] The high-resolution C 1s spectrum indicated the presence of C=C, C−C, C−S, C−N, and C=O bonds in the carbon framework, which originated from the carbon precursor polyvinylpyrrolidone K30 (PVP-K30) and sulfur source thioacetamide (TAA) (Figure 1c).[38–40]

Field-emission scanning electron microscopy (FESEM) images showed that $Fe_7S_8$@NC-PS precursor exhibited a porous hierarchical structure with an average diameter of 6.8 μm, which was constructed by interconnected 2D nanowalls (Figure 1d and Figure S3, Supporting Information). The thickness of the nanowall was around 47 nm (Figure 1e). After carbonization, the porous spherical structure was well maintained (Figure 1f). However, the diameter contracted to ≈5.6 μm with a reduced thickness around 30 nm (Figure 1g). Meanwhile, numerous $Fe_7S_8$ nanoparticles appeared on the surfaces of nanowalls, as indicated in the purple circles. The void space correspondingly increased owing to the shrinkage of nanowalls, which offered sufficient space to accommodate the large volume changes of $Fe_7S_8$ during cycling.

Transmission electron microscopy (TEM) image confirmed that the whole structure of $Fe_7S_8$@NC-PS was porous (**Figure 2**a). A high-resolution TEM (HRTEM) image showed that the edge of carbon nanowalls was amorphous phase (Figure 2b). Energy-dispersive X-ray spectroscopy (EDX) elemental mapping results (Figure 2c) presented the distribution of Fe, S, and C. The color contrast in Fe and S distribution further revealed a porous structure. The nitrogen adsorption–desorption isotherm exhibited that S-$Fe_7S_8$ had a Brunauer–Emmett–Teller (BET) surface area of 27.31 $m^2\ g^{-1}$ and total pore volume of 0.028 $cm^3\ g^{-1}$ (Figure S4, Supporting Information). Figure S5 (Supporting Information) shows that the pore size distribution centered around 2 nm with an average pore diameter of 3.264 nm. The large surface area provided $Fe_7S_8$@NC-PS sufficient contact area with electrolyte for fast reaction kinetics. Raman spectrum in Figure S6 (Supporting Information) further confirmed the existence of carbon. Thermogravimetric analysis (TGA) was carried out to determine the carbon ratio of $Fe_7S_8$@NC-PS (Figure 2d). The initial weight loss between 300 and 450 °C was attributed to the combustion of carbon


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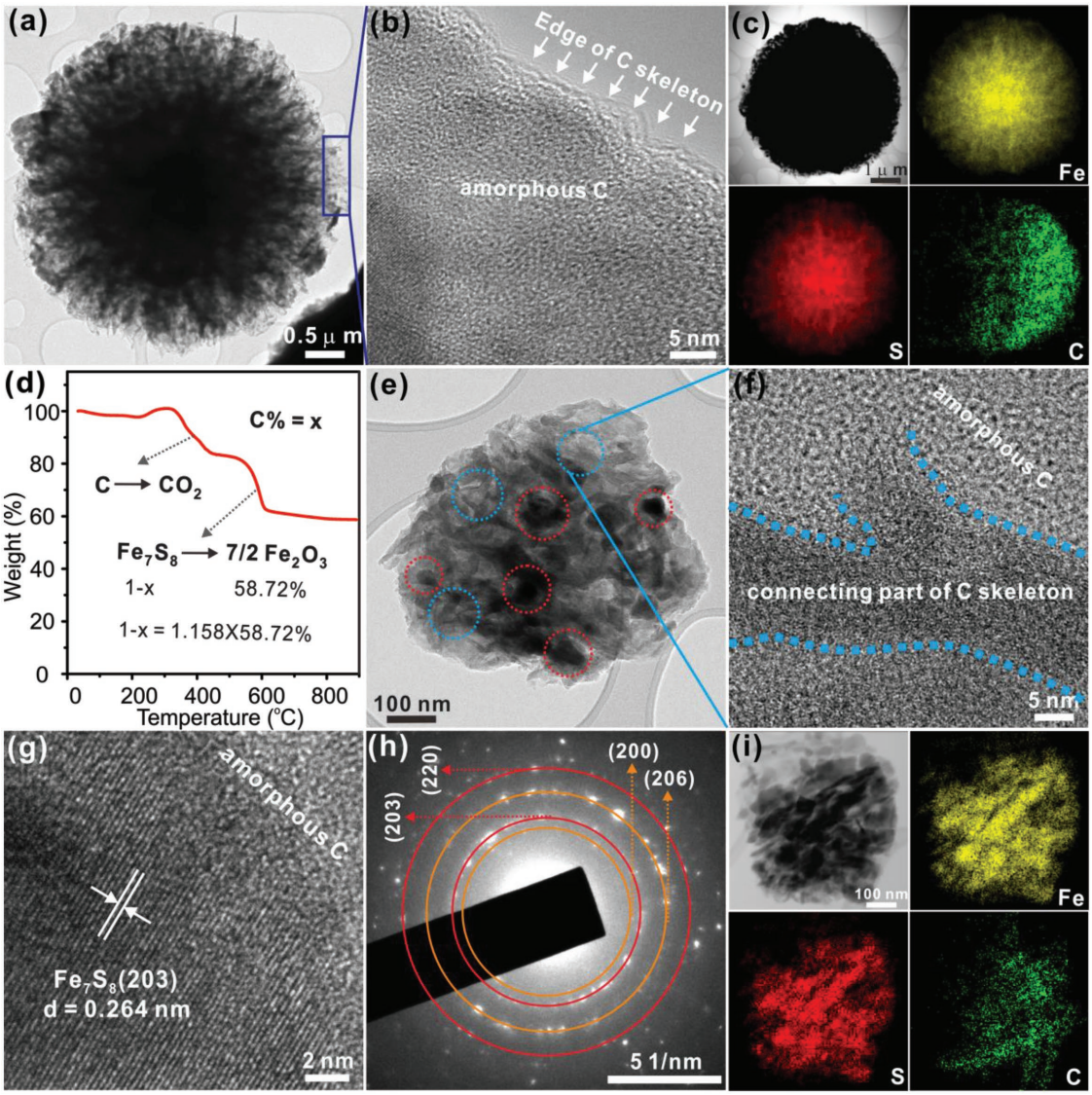


**Figure 2.** a) TEM image of $Fe_7S_8$@NC-PS. b) HRTEM image of the edge of $Fe_7S_8$@NC-PS. c) Elemental mapping images of $Fe_7S_8$@NC-PS. d) TGA analysis of $Fe_7S_8$@NC-PS. e) TEM image of the fragment of $Fe_7S_8$@NC-PS. f,g) HR-TEM images, h) SAED pattern, and i) elemental mapping images of the fragment of $Fe_7S_8$@NC-PS.

in air. The second weight loss between 500 and 610 °C was ascribed to the oxidation of $Fe_7S_8$ to $Fe_2O_3$.[41,42] By calculation, the carbon content in $Fe_7S_8$@NC-PS was 32%.

To well observe the internal morphology, $Fe_7S_8$@NC-PS was grinded into small fragments. Figure 2e shows that the fragment was composed of $Fe_7S_8$ nanoparticles with diameters ranging from 30 to 80 nm (marked by red circles), which were embedded in the carbon matrix. The cross sections of carbon nanowalls in blue circles demonstrated that the nanowall networks penetrated the whole structure. Figure 2f exhibits that the cross section of the connection part for carbon nanowall networks was still amorphous. HRTEM image displayed a set of parallel fringes with a *d*-spacing of 0.264 nm, corresponding to the (203) plane of the hexagonal $Fe_7S_8$ phase (Figure 2g). The selected area electron diffraction (SAED) pattern and the four diffraction rings corresponded to different lattice planes of $Fe_7S_8$ (Figure 2h). The elemental mapping images and corresponding EDX results of $Fe_7S_8$@NC-PS fragment further confirmed the structure and composition of S-$Fe_7S_8$ (Figure 2i and Figure S7, Supporting Information). All the results verify that an elaborate $Fe_7S_8$@NC-PS hierarchical structure has been successfully prepared.

Lithium storage properties of $Fe_7S_8$@NC-PS were evaluated in coin-type half-cells. **Figure 3**a shows the cyclic voltammetry (CV) curves for the first five cycles at a scan rate of 0.1 mV $s^{-1}$ between 0.02 and 2.5 V (vs $Li/Li^+$). During the first cathodic scan, a large peak appeared near 1.05 V, corresponding to $Li^+$ intercalation at high voltages to form $Li_xFeS_2$ ($x < 2$).[20,37,43] The other reduction peak around 0.68 V was assigned to the subsequent conversion reaction to form $Fe^0 + Na_2S$ and the formation of a SEI layer. In the anodic scan, one strong peak centered at 1.79 V corresponded to the desodiation reactions of $Fe^0 + Na_2S$


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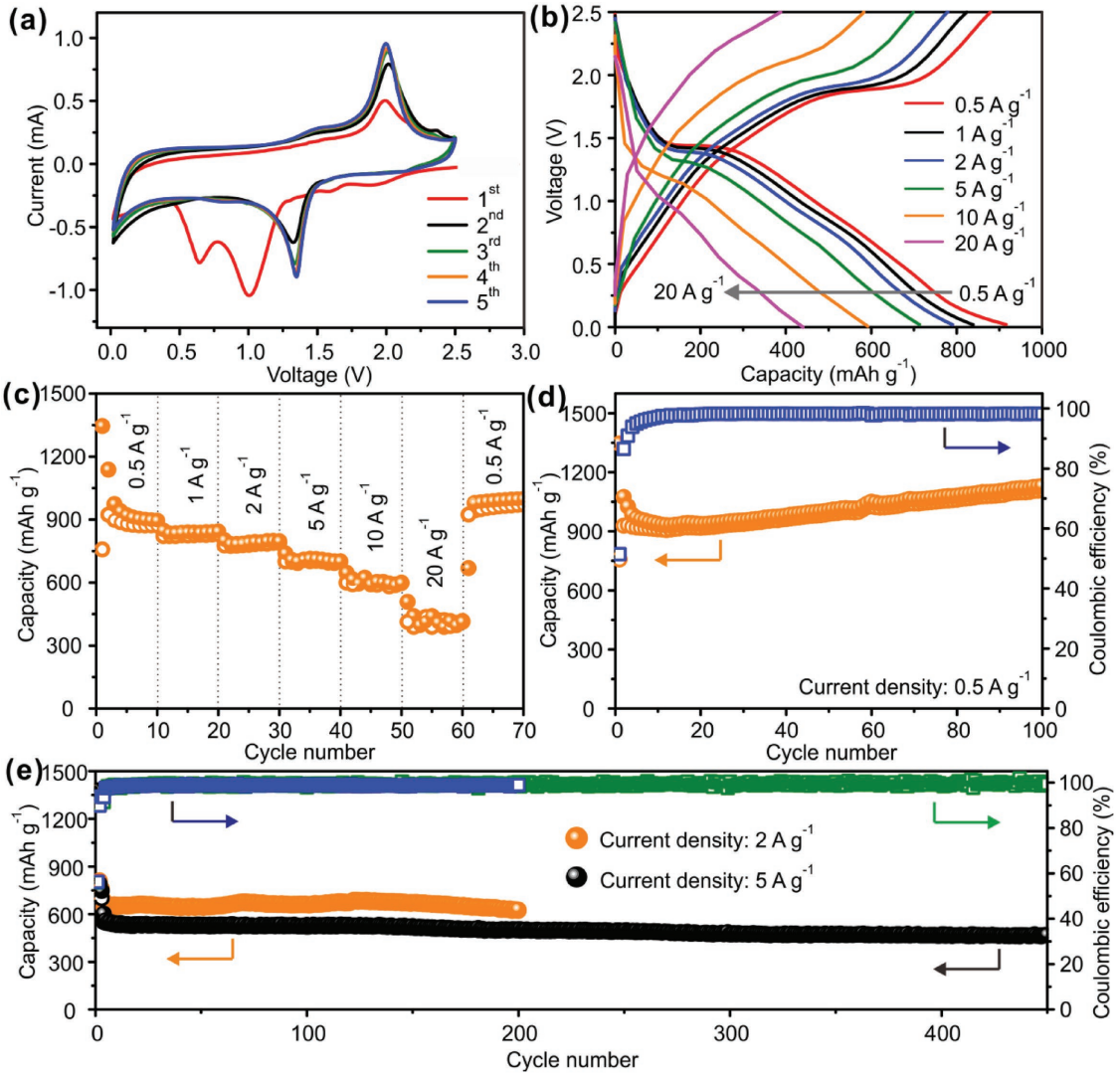


**Figure 3.** The electrochemical performance of $Fe_7S_8$@NC-PS in LIBs: a) CV curves of $Fe_7S_8$@NC-PS for the first five cycles at a scan rate of 0.1 mV $s^{-1}$. b,c) Voltage profiles and the corresponding rate performance of $Fe_7S_8$@NC-PS at various current densities from 0.5 to 20 A $g^{-1}$. Cycling performance of $Fe_7S_8$@NC-PS at the current densities of d) 0.5 A $g^{-1}$ and e) 2 and 5 A $g^{-1}$, respectively, with the corresponding Coulombic efficiencies.

to form $Na_{2-x}FeS_2$ ($0 < x < 2$). The irreversible formation of SEI layer was supposed to be the reason for the difference between the first scan and the subsequent scans for CV curves.[44,45] In the following cycles, the redox peaks were consistent with the reversible reactions between $Na_{2-x}FeS_2$ and $Fe^0$ + $Na_2S$.[20] Figure S8 (Supporting Information) shows the discharge and charge voltage profiles of $Fe_7S_8$@NC-PS at a current density of 0.5 A $g^{-1}$ with a voltage window of 0.02–2.5 V (vs $Li/Li^+$), which were well consistent with CV curves. The electrode gave rise to an initial discharge capacity of 1346 mAh $g^{-1}$ with Coulombic efficiencies (CE) around 60%. The irreversible capacity loss in the first cycle was mainly attributed to the trapped $Li^+$ ions, inevitable formation of SEI layer, and electrolyte decomposition.[45,46] After the first cycle, the specific capacities gradually stabilized around 930 mAh $g^{-1}$.

The rate performance of $Fe_7S_8$@NC-PS was evaluated at various current densities from 0.5 to 20 A $g^{-1}$ (Figure 3b,c). The average specific capacities were 920, 840, 790, 713, 600, and 440 mAh $g^{-1}$ at 0.5, 1, 2, 5, 10, and 20 A $g^{-1}$, respectively. Remarkably, a higher capacity than that of graphite was still maintained even at a high current rate of 20 A $g^{-1}$. When the current density went back to 0.5 A $g^{-1}$, $Fe_7S_8$@NC-PS electrode could quickly return to its original value, which was then kept very stable for extended cycles. The excellent rate capacities and stability benefited from the superior kinetics and structural stability of $Fe_7S_8$@NC-PS. The electrodes were further evaluated for long-term cycling stability using two different current densities. Figure 3d shows the cycling performance and corresponding CEs at 0.5 A $g^{-1}$ for 100 cycles. As can be seen, $Fe_7S_8$@NC-PS delivered a slightly increasing capacity, which was supposed to be the activation of the electrode materials.[47] The reversible formation/dissolution of organic polymeric gel-like layers by electrolyte decomposition was supposed to be another reason for delivering an extra capacity through a pseudo-capacitive behavior.[48,49] After 100 cycles, a high specific capacity of 1130 mAh $g^{-1}$ was


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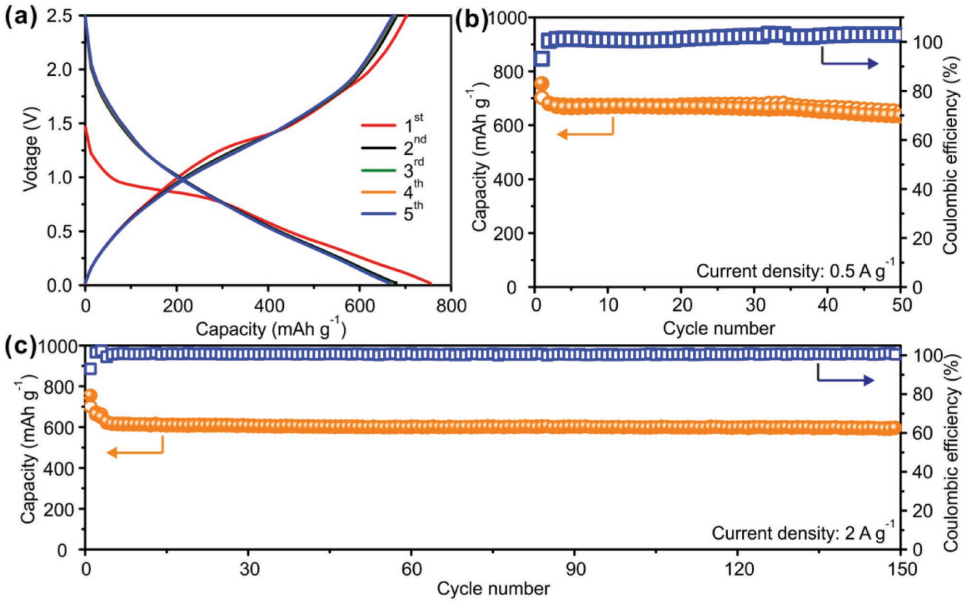


**Figure 4.** The electrochemical performance of $Fe_7S_8$@NC-PS in SIBs: a) Voltage profiles for the first five cycles at a current density of 0.5 A $g^{-1}$. b,c) Cycling performance at 0.5 and 2 A $g^{-1}$, respectively, and the corresponding Coulombic efficiencies.

retained. Meanwhile, the CE quickly increased to around 100% in the first ten cycles. In addition, $Fe_7S_8$@NC-PS demonstrated a long cycling stability when cycled at high current densities of 2 and 5 A $g^{-1}$ (Figure 3e), and the specific capacities stabilized around 630 and 470 mAh $g^{-1}$ after 200 and 450 cycles, respectively. More impressively, it delivered a reversible capacity of 438 mAh $g^{-1}$ after 700 cycles when cycled at a high current density of 10 A $g^{-1}$ (Figure S9, Supporting Information). The lithium storage properties of $Fe_7S_8$@NC-PS in terms of cycling stability and rate capability are comparable to the reported iron sulfide works as shown in Table S1 (Supporting Information).

Postmortem FESEM images show that the porous spherical structure still well maintained even after 100 cycles (Figure S10, Supporting Information), confirming the superior structural stability of $Fe_7S_8$@NC-PS. Electrochemical impedance spectroscopy (EIS) was applied to explore the charge transfer kinetics of $Fe_7S_8$@NC-PS during cycling (Figure S11, Supporting Information). As can be seen, the charge transfer resistance ($R_{ct}$) largely decreased after five cycles compared with the fresh cell, which could be attributed to the formation of a stable ion-conductive SEI layer and the activation of the electrode materials during cycling.[50–52] After 20 cycles, a well-maintained $R_{ct}$ further demonstrated the structural stability of $Fe_7S_8$@NC-PS.

$Fe_7S_8$@NC-PS was also tested as anode material for SIBs. Figure S12 (Supporting Information) shows CVs of the $Fe_7S_8$@NC-PS electrode at a voltage window of 0.02–2.5 V. A large cathodic peak centered at 0.82 V in the initial scan was assigned to the phase transition owing to the intercalation of $Na^+$ ions.[20,53] Another two peaks at 0.34 and 0.2 V were supposed to be the conversion reactions to further form $Na_2S$ and metallic Fe and the irreversible formation of SEI layer. In the following cycles, two main pairs of redox peaks at 0.8 and 0.41 V in cathodic scans and 1.32 and 1.73 V in anodic scans were ascribed to the multistep sodiation and desodiation reactions of $Fe_7S_8$@NC-PS, respectively. These well-overlapped CVs revealed a reversible sodiation and desodiation mechanism. The charge/discharge voltage profiles were well consistent with CV curves, as shown in **Figure 4**a. $Fe_7S_8$@NC-PS delivered a discharge capacity of 754 mAh $g^{-1}$ with a high initial CE of 93.2%. The specific capacities stabilized around 665 mAh $g^{-1}$ in the following five cycles. The cycling performance was evaluated at a current density of 0.5 A $g^{-1}$ (Figure 4b). As can be seen, the CEs quickly increased to 100% and the specific capacity remained at ≈650 mAh $g^{-1}$ after 50 cycles at 0.5 A $g^{-1}$. When evaluated at a high current density of 2A $g^{-1}$, $Fe_7S_8$@NC-PS exhibited an excellent cycling stability with a high specific capacity around 600 mAh $g^{-1}$ after 150 cycles (Figure 4c).

The superior electrochemical properties of $Fe_7S_8$@NC-PS in LIBs and SIBs benefited from its unique nano-micro architecture. This porous structure provided sufficient void space to accommodate the large volume changes during cycling, which enabled $Fe_7S_8$@NC-PS a good cycling stability. The interconnected nanowalls offered a large electrolyte/electrode contact area and short diffusion path for $Li^+$/$Na^+$ and electrons, which significantly improved the rate capability. In addition, the nitrogen-doped carbon framework contributed a high conductivity and structural stability upon charge and discharge, which further enhanced the cycling and rate performance.

In summary, a porous nano-micro hierarchical structure has been successfully prepared, which was composed of $Fe_7S_8$ nanoparticles embedded in nitrogen-doped carbon frameworks. With the merits in structure and composition, the $Fe_7S_8$@NC-PS electrodes demonstrated excellent electrochemical performance in LIBs and SIBs. Specifically, it delivered a stable reversible capacity of 630 mAh $g^{-1}$ after 200 cycles at 2A $g^{-1}$ and a high specific capacity of 440 mAh $g^{-1}$ even at 20 A $g^{-1}$ in LIBs. Impressively, it demonstrated an ultralong cycling stability with a reversible capacity of 438 mAh $g^{-1}$ after 700 cycles when cycled at a high current density of 10 A $g^{-1}$.


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In addition, $Fe_7S_8$@NC-PS revealed a good cycling stability in SIBs with a high capacity of 650 at 0.5 A $g^{-1}$ after 50 cycles and 600 mAh $g^{-1}$ at 2 A $g^{-1}$ after 150 cycles. These superior electrochemical properties enabled $Fe_7S_8$@NC-PS potential electrode materials for next-generation rechargeable batteries.

## Experimental Section

*Synthesis of $Fe_7S_8$@NC-PS*: In a typical synthesis, $FeCl_2 \cdot 4H_2O$ (2 mmol) was first dissolved in distilled water (DIW, 40 mL) to form a transparent solution. Afterward, polyvinylpyrrolidone K30 (PVP-K30, 0.15 g) and thioacetamide (5 mmol) were orderly added into the above solution under stirring for 30 min to form a homogeneous solution. The above solution was then transferred into a 50 mL Teflon-lined stainless steel autoclave and then kept at 200 °C for 12 h. After cooling down to room temperature naturally, the obtained dark-brown product was separated by centrifugation, washed with DIW and ethanol for several times before drying at 70 °C overnight under vacuum.

*Materials Characterization*: FESEM (JEOL JSM07600F) and TEM (JEOL JEM-2100F) were used to characterize the microscopic features of the samples. The composition of the samples and elemental mapping were analyzed by EDX spectroscopy attached to the TEM instrument. A Rigaku D/MAX RINT-2000 X-ray diffractometer with Cu K$\alpha$ radiation at a voltage of 40 kV and a current of 40 mA was used to collect the XRD patterns of the products. XPS (VG microtech ESCA2000) was used to investigate the surface composition and the valence states of the as-synthesized samples. Raman spectroscopy (JY LABRAM, $Ar^+$ laser excitation with a wavelength of 514.5 nm) was applied to check the existence of carbon. The specific surface areas and pore size distribution of as-synthesized materials were measured using BET method on Autosorb-1 at liquid-nitrogen temperature.

*Electrochemical Measurements*: The battery tests were carried out in a half-cell configuration. The working electrode consists of active materials, conductivity agent (Super P), and polymer binder (polyvinylidene fluoride, PVDF) with a weight ratio of 70:20:10. The active mass loading on the electrode is about 1.2 mg $cm^{-2}$. The electrolytes were a solution of 1.3 M $LiPF_6$ in a mixture of ethylene carbonate and diethyl carbonate (EC/DEC, 1:1 by wt%) with 5% fluoroethylene carbonate (FEC) additive and 1 M $NaSO_3CF_3$ in diethylene glycol dimethyl ether (DEGDME) for LIBs and SIBs, respectively. Lithium and sodium discs were used as both counter electrodes and reference electrodes. The coin-type half cells were assembled in argon-filled glove box and then tested in TOSCAT 3000 battery tester (TOSCAT 3000, Toyo Systems, Tokyo, Japan) with a voltage range between 0.02 and 2.5 V for LIBs and SIBs. CV curves were tested using AUTOLAB potentiostat/galvanostat apparatus (AUT85698) with a scan rate of 0.1 mV $s^{-1}$. The EIS data were tested with a frequency range from 100 kHz to 100 mHz.

## Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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## Conflict of Interest

The authors declare no conflict of interest.

## Keywords

batteries, $Fe_7S_8$, nano-micro hybrid, nitrogen doped, porous carbon framework

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